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(54) **CHEMICAL STRUCTURE GENERATING DEVICE, CHEMICAL STRUCTURE GENERATING PROGRAM, AND CHEMICAL STRUCTURE GENERATING METHOD**

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G16C 20/70 (2019.01)

(52) **U.S. Cl.**
CPC ***G16C 20/10*** (2019.02); ***G16C 20/70*** (2019.02)

(58) **Field of Classification Search**
CPC G16C 20/10; G16C 20/70
USPC 703/2
See application file for complete search history.

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(57) **ABSTRACT**

A chemical structure generating device according to the present invention includes a generator and a controller. The generator produces a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list. The controller applies the product list as a new reactant list to the generator, updates a database having at least one list of the reactant list and the product list, and allows the generator to produce a new product list based on the new reactant list and the chemical reaction list.

14 Claims, 6 Drawing Sheets

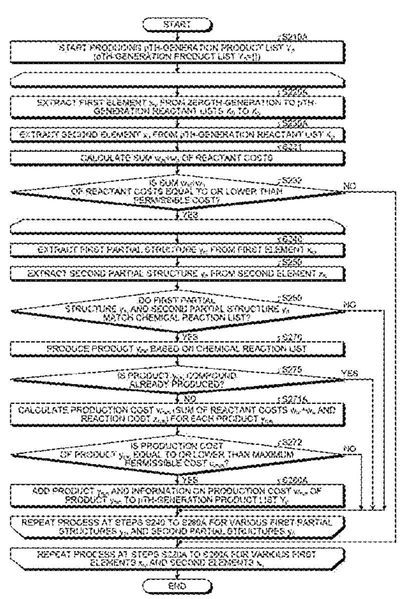

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FIG.1

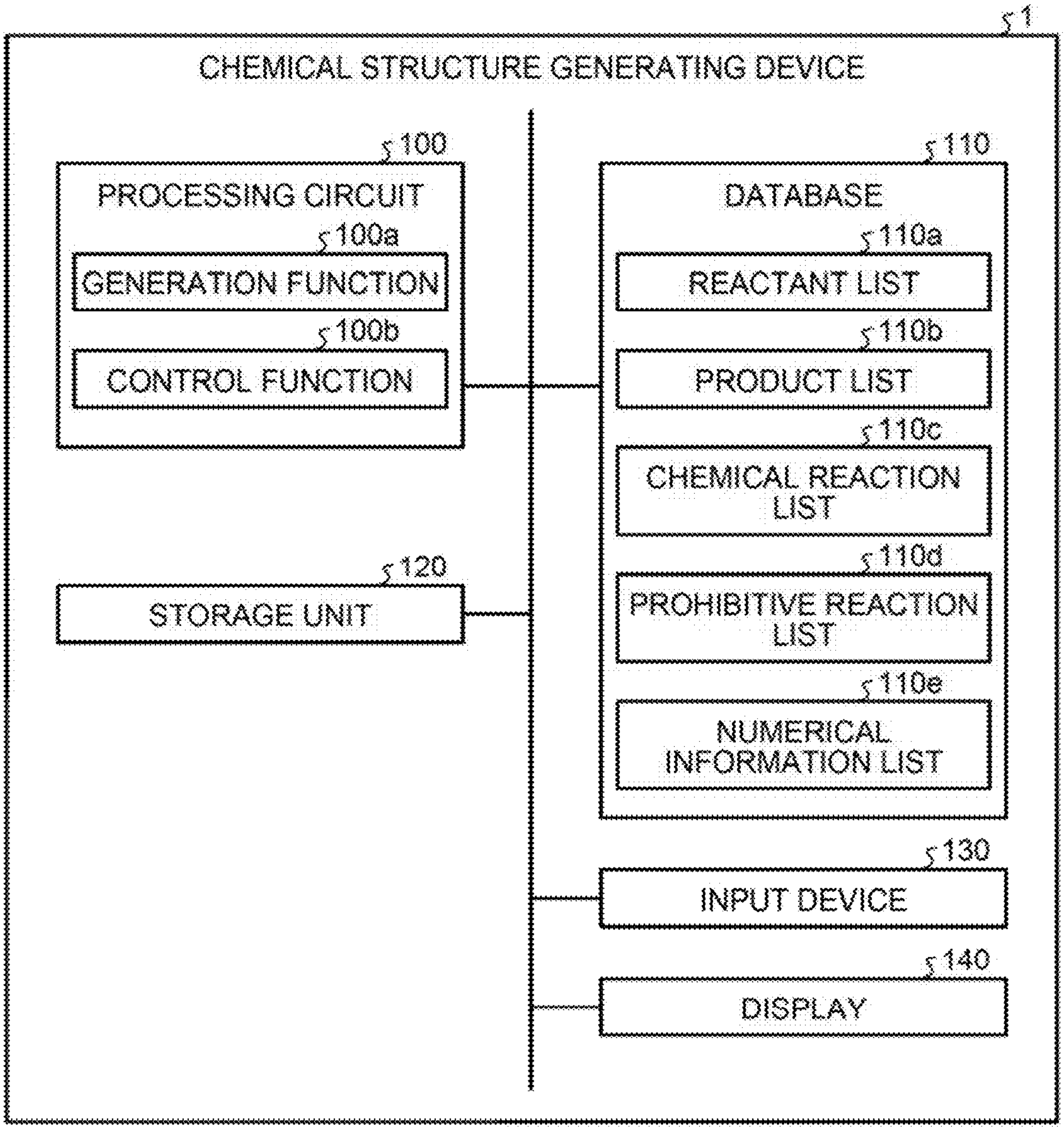


FIG.2

MOLECULAR STRUCTURE 1	MOLECULAR STRUCTURE 2	GENERATED STRUCTURE
X_1-COOH	X_2-H	X_1-CO-X_2
X_1-CO-X_2		X_1-CHOH-X_2
X_1-CHOH-X_2	X_3-H	X_1-CHX_3-X_2

FIG.3

MOLECULAR STRUCTURE 1	MOLECULAR STRUCTURE 2	GENERATED STRUCTURE	Rule
X_1-COOH	X_2-H	X_1-CO-X_2	Allow
X_1-COOH	X_2-CO-CH_2-CO-X_3	X_2-CO-CHX_1-CO-X_3	Deny

FIG.4

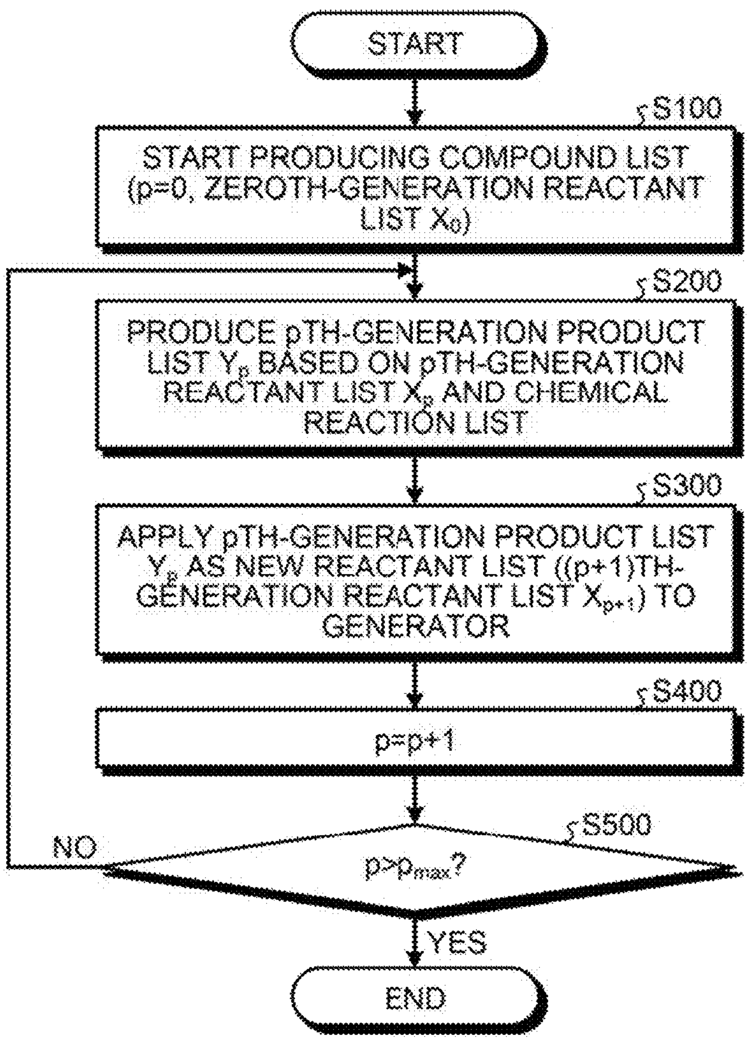


FIG.5

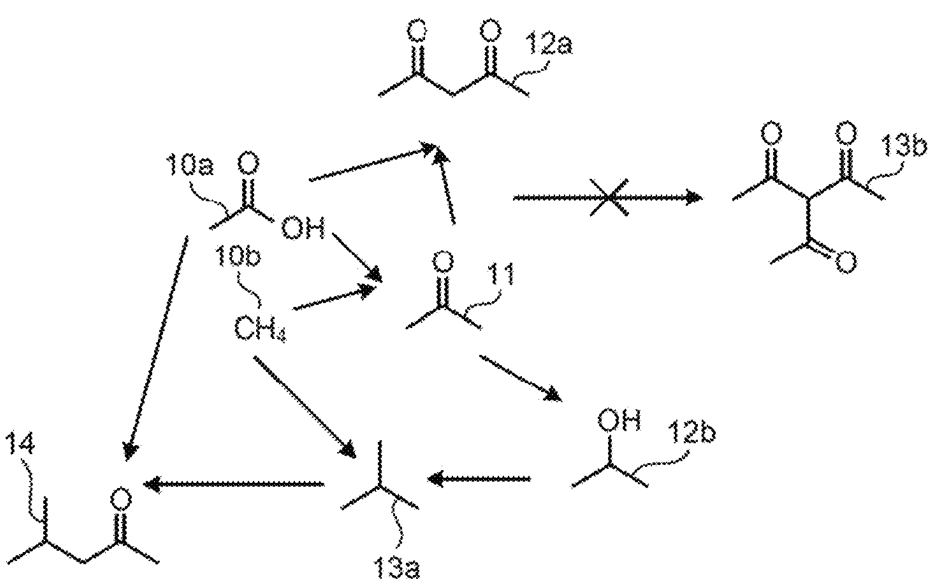


FIG.6

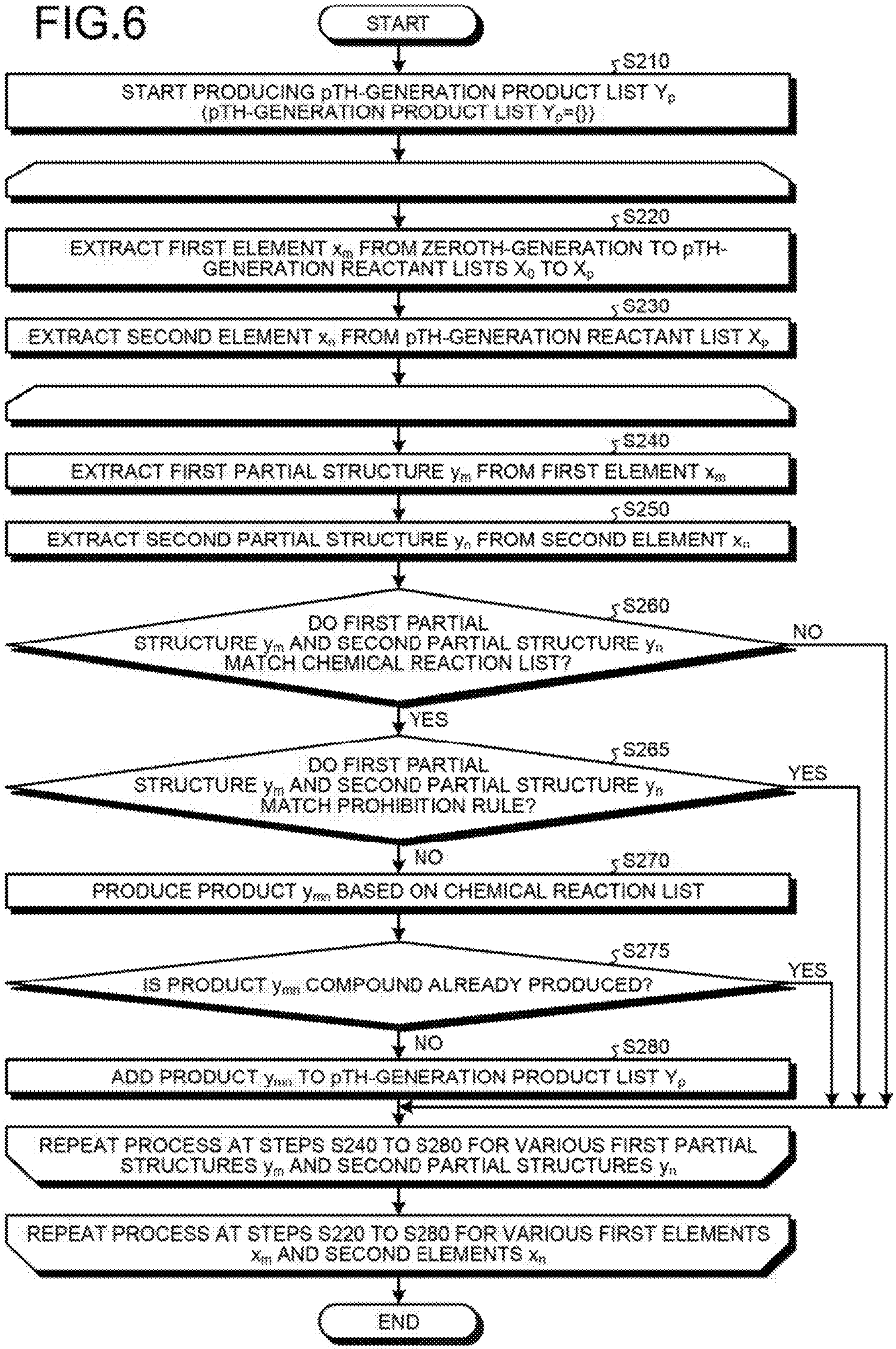
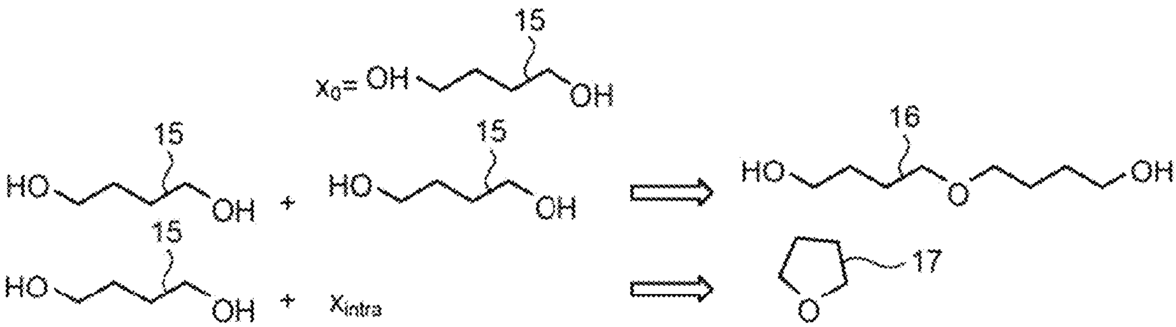
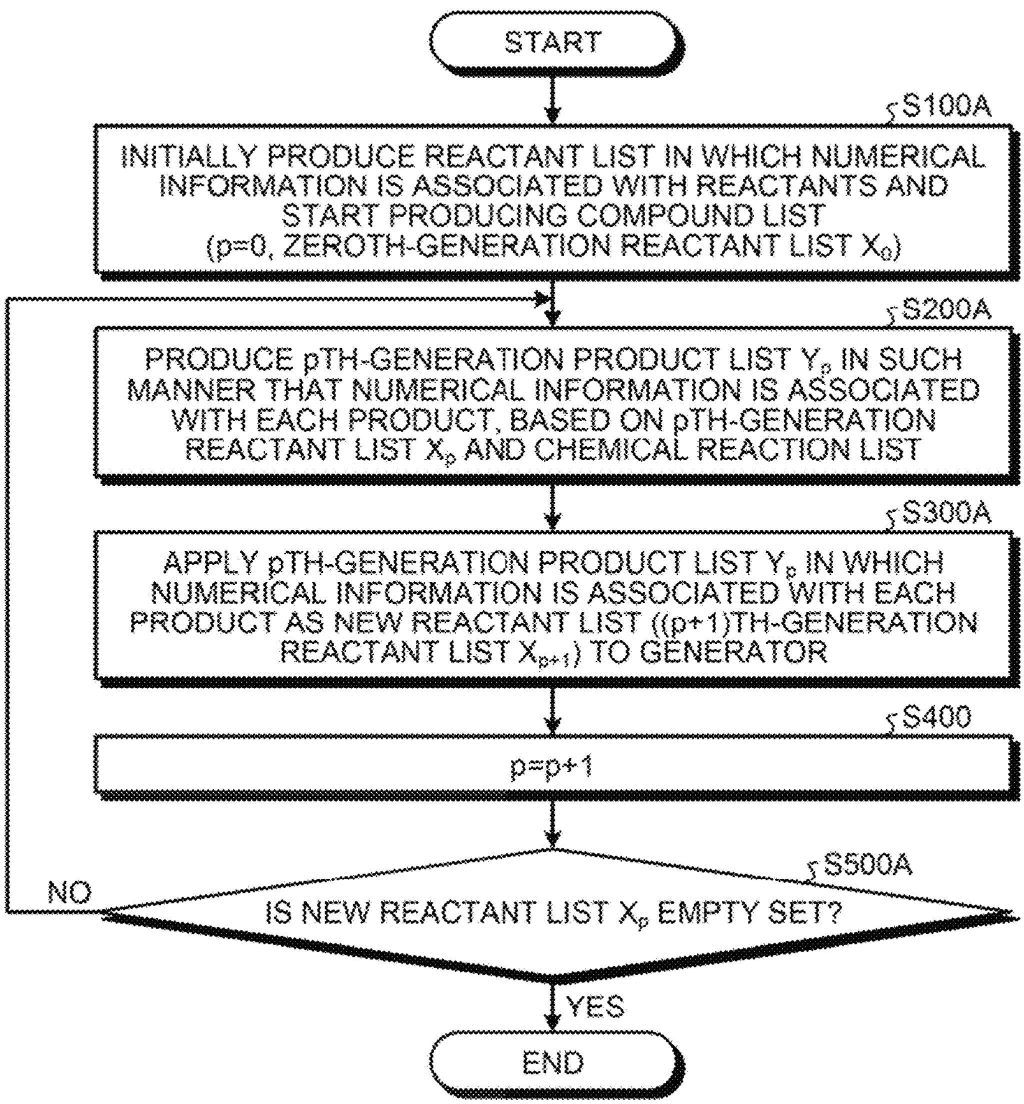

START
S210
START PRODUCING pTH-GENERATION PRODUCT LIST Y_p (pTH-GENERATION PRODUCT LIST Y_p={})
S220
EXTRACT FIRST ELEMENT x_m FROM ZEROTH-GENERATION TO pTH-GENERATION REACTANT LISTS X_0 TO X_p
S230
EXTRACT SECOND ELEMENT x_n FROM pTH-GENERATION REACTANT LIST X_p
S240
EXTRACT FIRST PARTIAL STRUCTURE y_m FROM FIRST ELEMENT x_m
S250
EXTRACT SECOND PARTIAL STRUCTURE y_n FROM SECOND ELEMENT x_n
S260
DO FIRST PARTIAL STRUCTURE y_m AND SECOND PARTIAL STRUCTURE y_n MATCH CHEMICAL REACTION LIST?
NO
YES
S265
DO FIRST PARTIAL STRUCTURE y_m AND SECOND PARTIAL STRUCTURE y_n MATCH PROHIBITION RULE?
YES
NO
S270
PRODUCE PRODUCT y_{mn} BASED ON CHEMICAL REACTION LIST
S275
IS PRODUCT y_{mn} COMPOUND ALREADY PRODUCED?
YES
NO
S280
ADD PRODUCT y_{mn} TO pTH-GENERATION PRODUCT LIST Y_p
REPEAT PROCESS AT STEPS S240 TO S280 FOR VARIOUS FIRST PARTIAL STRUCTURES y_m AND SECOND PARTIAL STRUCTURES y_n
REPEAT PROCESS AT STEPS S220 TO S280 FOR VARIOUS FIRST ELEMENTS x_m AND SECOND ELEMENTS x_n
END

FIG.7



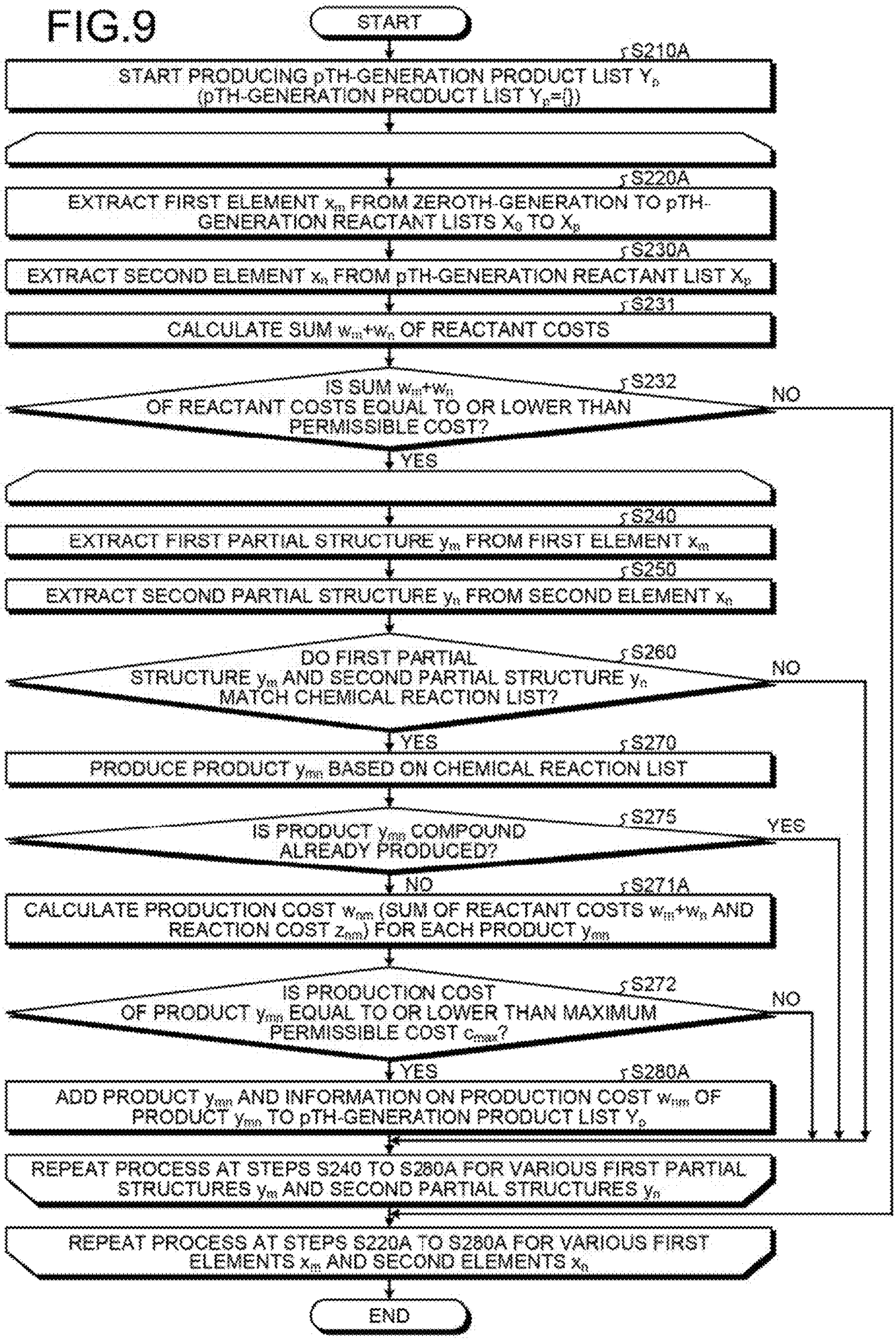
X_0= OH
OH
15
HO
OH
+
15
HO
OH
16
HO
O
OH
15
HO
OH
+
X_{intra}
17
O

FIG.8



START
S100A
INITIALLY PRODUCE REACTANT LIST IN WHICH NUMERICAL INFORMATION IS ASSOCIATED WITH REACTANTS AND START PRODUCING COMPOUND LIST (p=0, ZEROTH-GENERATION REACTANT LIST X_0)
S200A
PRODUCE pTH-GENERATION PRODUCT LIST Y_p IN SUCH MANNER THAT NUMERICAL INFORMATION IS ASSOCIATED WITH EACH PRODUCT, BASED ON pTH-GENERATION REACTANT LIST X_p AND CHEMICAL REACTION LIST
S300A
APPLY pTH-GENERATION PRODUCT LIST Y_p IN WHICH NUMERICAL INFORMATION IS ASSOCIATED WITH EACH PRODUCT AS NEW REACTANT LIST ((p+1)TH-GENERATION REACTANT LIST X_{p+1}) TO GENERATOR
S400
p=p+1
S500A
IS NEW REACTANT LIST X_p EMPTY SET?
NO
YES
END

FIG.9


START
S210A
START PRODUCING pTH-GENERATION PRODUCT LIST Y_p (pTH-GENERATION PRODUCT LIST Y_p={})
S220A
EXTRACT FIRST ELEMENT x_m FROM ZEROTH-GENERATION TO pTH-GENERATION REACTANT LISTS X_0 TO X_p
S230A
EXTRACT SECOND ELEMENT x_n FROM pTH-GENERATION REACTANT LIST X_p
S231
CALCULATE SUM w_m+w_n OF REACTANT COSTS
S232
IS SUM w_m+w_n OF REACTANT COSTS EQUAL TO OR LOWER THAN PERMISSIBLE COST?
NO
YES
S240
EXTRACT FIRST PARTIAL STRUCTURE y_m FROM FIRST ELEMENT x_m
S250
EXTRACT SECOND PARTIAL STRUCTURE y_n FROM SECOND ELEMENT x_n
S260
DO FIRST PARTIAL STRUCTURE y_m AND SECOND PARTIAL STRUCTURE y_n MATCH CHEMICAL REACTION LIST?
NO
YES
S270
PRODUCE PRODUCT y_{mn} BASED ON CHEMICAL REACTION LIST
S275
IS PRODUCT y_{mn} COMPOUND ALREADY PRODUCED?
YES
NO
S271A
CALCULATE PRODUCTION COST w_{nm} (SUM OF REACTANT COSTS w_m+w_n AND REACTION COST z_{nm}) FOR EACH PRODUCT y_{mn}
S272
IS PRODUCTION COST OF PRODUCT y_{mn} EQUAL TO OR LOWER THAN MAXIMUM PERMISSIBLE COST c_{max}?
NO
YES
S280A
ADD PRODUCT y_{mn} AND INFORMATION ON PRODUCTION COST w_{nm} OF PRODUCT y_{mn} TO pTH-GENERATION PRODUCT LIST Y_p
REPEAT PROCESS AT STEPS S240 TO S280A FOR VARIOUS FIRST PARTIAL STRUCTURES y_m AND SECOND PARTIAL STRUCTURES y_n
REPEAT PROCESS AT STEPS S220A TO S280A FOR VARIOUS FIRST ELEMENTS x_m AND SECOND ELEMENTS x_n
END

CHEMICAL STRUCTURE GENERATING DEVICE, CHEMICAL STRUCTURE GENERATING PROGRAM, AND CHEMICAL STRUCTURE GENERATING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of PCT International Application No. PCT/JP2020/031178 filed on Aug. 18, 2020 which claims the benefit of priority from Japanese Patent Application No. 2019-149873 filed on Aug. 19, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a chemical structure generating device, a chemical structure generating program, and a chemical structure generating method.

2. Description of the Related Art

There is interest in methods to design molecules having useful properties by machine learning using computer simulations. As an example, first, in a learning phase, a trained model is produced by machine learning such as deep learning using a training dataset, and then in an execution phase, for example, prediction of physical property values or search for molecules having desired physical properties is performed using the trained model produced in the learning phase.

Since performing machine learning such as deep learning requires a large volume of training data, how high-quality training data is efficiently produced is important. In view of this, a chemical structure generating device that automatically generates molecular structures appropriate for use as training data in machine learning is important.

Here, as the number of atoms increases, the number of possible molecular structures for the number of atoms exponentially increases to cause a combinatorial explosion. It is therefore actually impossible to exhaustively generate molecular structure candidates. In addition, the ratio of the number of usable molecules, that is, molecules actually stably existing and commercially usable to the number of possible molecular structures rapidly decreases as the number of atoms increases. In performing machine learning, it is extremely inefficient in terms of time and economy to use training data including many molecular structures commercially unusable. There is therefore a demand for a chemical structure generating device capable of efficiently generating only molecules commercially usable and valuable. Conventional technologies are described in WO 9501606, United States Patent Application No. 2018/096100, Japanese Patent Application Laid-open No. 2001-058962, and WO 9736252, for example.

SUMMARY OF THE INVENTION

It is an object of the present invention to at least partially solve the problems in the conventional technology.

It is an object of the present invention to efficiently generate molecular structures suitable for use as training data for machine learning.

In order to solve the problems and to achieve the object, a chemical structure generating device of the present invention comprises a generator and a controller. The generator produces a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list. The controller applies the product list as a new reactant list to the generator, update a database having at least one list of the reactant list and the product list, and allow the generator to produce a new product list based on the new reactant list and the chemical reaction list.

Further, a chemical structure generating program of the present invention causes a computer to perform:

- a process of producing a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list;
- a process of applying the product list as a new reactant list;
- a process of updating a database having at least one list of the reactant list and the product list; and
- a process of producing a new product list based on the new reactant list and the chemical reaction list.

Further, a chemical structure generating method of the present invention is a method performed by a chemical structure generating device, the chemical structure generating method comprising:

- producing a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list;
- applying the product list as a new reactant list;
- updating a database having at least one list of the reactant list and the product list; and
- producing a new product list based on the new reactant list and the chemical reaction list.

According to the present invention, it is possible to efficiently generate molecular structures suitable for use as training data for machine learning.

The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram illustrating an overview of a chemical structure generating device according to embodiments of the present invention;

FIG. **2** is a diagram illustrating an example of a chemical reaction list in the chemical structure generating device according to a first embodiment;

FIG. **3** is a diagram illustrating an example of a prohibitive reaction list in the chemical structure generating device according to the first embodiment;

FIG. **4** is a flowchart illustrating a process performed by the chemical structure generating device according to the first embodiment;

FIG. **5** is a diagram illustrating a process performed by the chemical structure generating device according to the first embodiment;

FIG. **6** is a flowchart illustrating the detail of the process at step **S200** in FIG. **4**;

FIG. **7** is a diagram illustrating a process performed by the chemical structure generating device according to the first embodiment;

FIG. **8** is a flowchart illustrating a process performed by the chemical structure generating device according to a second embodiment; and

FIG. **9** is a flowchart illustrating the detail of the process at step S**200**A in FIG. **8**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

A chemical structure generating device according to embodiments of the present invention will be described below with reference to the accompanying drawings.

First Embodiment

FIG. **1** is a diagram illustrating an overview of a chemical structure generating device **1** according to embodiments of the present invention. The chemical structure generating device **1** includes a processing circuit **100**, a storage unit **120**, a database **110**, an input device **130**, and a display **140**. The processing circuit **100** includes a generation function **100***a* and a control function **100***b*. The database **110** includes a reactant list **110***a* that is a list of reactants and the like of chemical reactions, a product list **110***b* that is a list of products and the like of chemical reactions, a chemical reaction list **110***c*, a prohibitive reaction list **110***d*, and a numerical information list **110***e*. The chemical structure generating device **1** is a device for generating chemical structures serving as a basis for training data for use in machine learning, in the form of the reactant list **110***a* or the product list **110***b*.

The processing circuit **100** is a processor (for example, a central processing unit (CPU), a graphical processing unit (GPU), an application specific integrated circuit (ASIC), or a programmable logic device (PLD)) that reads and executes a computer program from the storage unit **120** to implement functions such as the generation function **100***a* and the control function **100***b*. That is, the processing functions including the generation function **100***a* and the control function **100***b* are stored in the storage unit **120** in the form of a computer program executable by the processor. The processing circuit **100** in a state in which the computer program has been read has functions such as the generation function **100***a* and the control function **100***b*. In other words, the generation function **100***a* and the control function **100***b* are an example of the generator and the controller, respectively. The detail of the process in each function of the processing circuit **100** will be described later.

The database **110** is a variety of databases referred to by the processing circuit **100** in a process of generating new compounds by the chemical structure generating device **1** and is configured with, for example, the reactant list **110***a*, the product list **110***b*, the chemical reaction list **110***c*, the prohibitive reaction list **110***d*, and the numerical information list **110***e*. These databases are stored in the storage unit **120** if necessary.

The chemical reaction list **110***c*, the prohibitive reaction list **110***d*, and the numerical information list **110***e* are data used by the processing circuit **100** to perform the process of generating chemical structures. On the other hand, the reactant list **110***a* or the product list **110***b* is data serving as a basis for training data used in machine learning at a subsequent stage, and producing these lists is one of the objects of the process performed by the chemical structure generating device **1**.

The reactant list **110***a* is a list of reactants serving as a basis for chemical reactions and includes compounds representing the reactants. More precisely, the reactant list **110***a* includes one or more elements, and each of the elements is a single element, a compound, an element representing an operation of not selecting a reactant, or an element representing an intramolecular reaction. Here, the chemical structure generating device **1** according to the embodiments successively updates the reactant list, starting from an initial compound. The reactant list **110***a* therefore includes an initial-state (zeroth-generation (p=0)) reactant list X_0, a first-generation reactant list X_1, a second-generation reactant list X_2, . . . , and a pth-generation reactant list X_p. Here, a (k+1)th generation reactant list X_{k+1} may completely include a kth-generation reactant list X_k, or the (k+1)th generation reactant list X_{k+1} need not completely include the kth-generation reactant list X_k.

The reactant list X_k of each generation may include special compounds x_{null} and x_{intra} indicating a special operation as optional constituent elements, in addition to compounds $x_{k1} \ldots x_{kn}$ representing reactants. These special compounds will be described later. When these special compounds are not included as elements, the elements of the kth-generation reactant list X_k are compounds x_{k1}, x_{k2} . . . , x_{kn} representing kth-generation reactants. Conversely, when these special compounds are included as elements, the elements of the kth-generation reactant list X_k are, for example, x_{k1}, x_{k2} . . . , x_{kn}, x_{null}, x_{intra}.

The product list **110***b* is a product list of chemical reactions and includes compounds representing products. More precisely, the product list **110***b* includes one or more elements, and each of the elements is a single element, a compound, an element representing an operation of not selecting a reactant, or an element representing an intramolecular reaction. Here, the product list **110***b* includes a zeroth-generation (p=0) product list Y_0, a first-generation product list Y_1, a second-generation product list Y_2, . . . , and a pth-generation product list Y_p, in the same manner as the reactant list **110***a*.

In some embodiments, the elements of the (k+1)th-generation reactant list X_{k+1} are equal to the elements of the kth-generation product list Y_k.

The chemical reaction list **110***c* is a possible chemical reaction list and is a list in which a list of partial structures of a reactant is associated with a list of partial structures of a product produced by a chemical reaction using the reactant. The chemical reaction list **110***c* also includes information on the cost of the chemical reaction, if necessary.

FIG. **2** illustrates an example of the chemical reaction list **110***c* with two or one reactant and one product corresponding thereto.

The first row of FIG. **2** provides a case where a structure "X_1—CO—X_2" is generated from a partial structure "X_1—COOH" and a partial structure "X_2—H". In such a case, a list {{"X_1—CO—X_2"}, {"X_1—COOH", "X_2—H"}} in which a partial structure list of reactants {"X_1—COOH", "X_2—H"} is associated with a partial structure list of a product {"X_1—CO—X_2"} produced by the chemical reaction is an example of the chemical reaction list **110***c*.

The second row of FIG. **2** provides a case where a structure "X_1—CHOH—X_2" is generated from a partial structure "X_1—CO—X_2". In such a case, a list {{"X_1—CHOH—X_2"}, {"X_1—CO—X_2"}} in which a partial structure list of a reactant {"X_1—CO—X_2"} is associated with a partial structure list of a product ("X_1—CHOH—X_2") produced by the chemical reaction is an example of the chemical reaction list **110***c*. Assuming that H_2 is also a partial structure, this chemical reaction may be represented by {{"X_1—CHOH—X_2"}, {"X_1—CO—X_2", "H_2"}}.

The third row of FIG. 2 provides a case, for example, where a structure "X_1—CHX_3—X_2" is generated from a partial structure "X_1—CHOH—X_2" and a partial structure "X_3—H". In such a case, a list {{"X_1—CHX_3—X_2"}, {"X_1—CHOH—X_2", "X_3—H"}} in which a partial structure list of reactants {"X_1—CHOH—X_2", "X_3—H"} is associated with a partial structure list of a product ("X_1—CHX_3—X_2") produced by the chemical reaction is an example of the chemical reaction list **110***c*.

FIG. **2** illustrates the cases with two or one reactant and one product corresponding thereto. However, the embodiments are not limited thereto, and the number of reactants may be three or more. For example, when a structure D is generated from a partial structure A, a partial structure B, and a partial structure C, a list {{D}, {A,B,C}} in which a partial structure list of reactants {A,B,C} is associated with a partial structure list of a product {D} produced by the chemical reaction is an example of the chemical reaction list **110***c*.

In the cases described above, the number of products is one. However, the embodiments are not limited thereto and the number of products may be two or more. For example, as a reverse reaction of the first row of FIG. **2**, a case where a structure "X_1—COOH" and a structure "X_2—H" are generated from a partial structure "X_1—CO—X_2" will be discussed. In such a case, a list {{"X_1—COOH", "X_2—H"}, {"X_1—CO—X_2"}} in which a partial structure list of a reactant {"X_1—CO—X_2"} is associated with a partial structure list of products {"X_1—COOH", "X_1—H"} produced by the chemical reaction is an example of the chemical reaction list **110***c*.

What is described above is represented by a formula as follows. For example, when a new structure y_{kl}' is produced from a partial structure y_k and a partial structure y_l, a list $\{\{y_{kl}'\}, \{y_k, y_l\}\}$ is the chemical reaction list **110***c*. In general, when new m structures $y_{q1q2\ldots qn;1}, y_{q1q2\ldots qn;2}, \ldots, y_{q1q2\ldots qn;m}$ are produced from n partial structures $y_{q1}, y_{q2}\ldots y_{qn}$, a list $\{\{y_{q1q2\ldots qn;1}, y_{q1q2\ldots qn;2}, \ldots, y_{q1q2\ldots qn;m}\}, \{y_{q1}, y_{q2}\ldots y_{qn}\}\}$ is the chemical reaction list **110***c*.

In some embodiments, unlike the reactant list **110***a* or the product list **110***b*, the elements of the chemical reaction list **110***c* are not updated for each generation, and the elements are fixed throughout generations.

The prohibitive reaction list **110***d* is a list of prohibited chemical reactions and is a list in which a partial structure list of reactants and a partial structure list of products produced by a chemical reaction using the reactant(s) are associated together with information that it is a prohibited chemical reaction.

For example, when a new structure y_{kl}' is produced from a partial structure y_k and a partial structure y_l, if the new structure to be produced has no commercial value and is not appropriate as a product, a list $\{\{y_{kl}'\}, \{y_k, y_l\}\}$ is the prohibitive reaction list **110***d*. In general, when new m structures $y_{q1q2\ldots qn;1}, y_{q1q2\ldots qn;2}, \ldots, y_{q1q2\ldots qn;m}$ are produced from n partial structures $q_{q1}, q_{q2}\ldots q_{qn}$, if the product is not appropriate as a product for the reasons such as no commercial value of the product, a list $\{\{y_{q1q2\ldots qn;1}, y_{q1q2\ldots qn;2}, \ldots, q_{q1q2\ldots qn;m}\}, \{y_{q1}, y_{q2}\ldots y_{qn}\}\}$ is the prohibitive reaction list **110***d*. The presence or absence of commercial value is determined depending on the fields, for example, in terms of cost, stability, toxicity, and the like.

FIG. **3** illustrates an example of the prohibitive reaction list **110***d*. FIG. **3** is a diagram for explaining a process related to the prohibitive reaction list **110***d*.

The upper section of FIG. **3** provides a case where a structure "X_1—CO—X_2" is generated from a partial structure "X_1—COOH" and a partial structure "X_2—H". Since there is no particular reason to limit generation of such a structure, generation of this structure is allowed (Allow).

On the other hand, the lower section of FIG. **3** provides a case where a structure "X_x—CO—CHX_1—CO—X_3" is generated from a partial structure "X_1—COOH" and a partial structure "X_2—CO—CH_2—CO—X_3". When it is determined that the produced structure is chemically unstable and has no commercial value in a certain field, for example, generation of such a structure is prohibited, and generation of this structure is denied (Deny). Specifically, a list {{"X_2—CO—CHX_1—CO—X_3"}, {"X_1—COOH", "X_2—CO—CH_2—CO—X_3"}} is the prohibitive reaction list **110***d*.

The numerical information list **110***e* is a numerical information list representing the cost and the like of each compound in the reactant list **110***a*. The numerical information list **110***e* will be described in detail in a second embodiment.

The storage unit **120** is a storage area such as a random access memory (RAM), a read-only memory (ROM), a flash memory, or a hard disk and stores a variety of computer programs to be executed by the processing circuit **100**, the execution results of the computer programs, the database **110**, and the like.

The input device **130** is a device for a user using the chemical structure generating device **1** to perform a variety of operations. The input device **130** is configured with, for example, a mouse, a keyboard, a touch panel, or a hardware keypad.

The display **140** shows a variety of information. For example, the display **140** shows a processing result of a CPU and a graphical user interface (GUI) for accepting a variety of operations from the user. The display **140** is configured with, for example, a liquid crystal display, an organic electro-luminescence (EL) display, or a cathode-ray tube display. The input device **130** and the display **140** may be integrated, for example, in a form such as a touch panel.

The embodiments are not limited to the foregoing examples. In some embodiments, the storage unit **120**, the input device **130**, the display **140**, and the like are not essential components of the chemical structure generating device **1**, that is, the compound generating device **1** need not include the storage unit **120**, the input device **130**, or the display **140**. As another example, the storage unit **120**, the input device **130**, the display **140**, and the like may be arranged outside the compound generating device **1**, for example, through a network and may exchange data with the processing circuit **100**.

The background of the embodiment will now be described.

There is interest in methods to design molecules having useful properties by machine learning using computer simulations. As an example, first, in a learning phase, a trained model is produced by machine learning such as deep learning using a training dataset, and then in an execution phase, for example, prediction of physical property values or search for molecules having desired physical properties is performed using the trained model produced in the learning phase.

Here, since performing machine learning such as deep learning requires a large volume of training data, how high-quality training data is efficiently produced is important. In view of this, a chemical structure generating device that automatically generates molecular structures appropriate for use as training data in machine learning is important.

Here, as the number of atoms increases, the number of possible molecular structures for the number of atoms exponentially increases to cause a combinatorial explosion. It is therefore actually impossible to exhaustively generate molecular structure candidates. In addition, the ratio of the number of usable molecules, that is, molecules actually stably existing and commercially usable to the number of possible molecular structures rapidly decreases as the number of atoms increases. In performing machine learning, it is extremely inefficient in terms of time and economy to use training data including many molecular structures commercially unusable. There is therefore a demand for a chemical structure generating device capable of efficiently generating only molecules commercially usable and valuable.

In view of such a background, the processing circuit **100** of the chemical structure generating device **1** according to the embodiments includes the generation function **100***a* and the control function **100***b*. The generation function **100***a* of the processing circuit **100** produces the product list **110***b* that is a list of products including one or more compounds, based on the reactant list **110***a* that is a list of reactants including one or more compounds and the chemical reaction list **110***c* that is a list of chemical reactions. The control function **100***b* of the processing circuit **100** applies the product list **110***b* as a new reactant list to a computer program related to the generation function **100***a*, updates the database **110** having at least one list of the reactant list **110***a* and the product list **110***b*, and causes the computer program related to the generation function **100***a* to produce a new product list based on the new reactant list and the chemical reaction list **110***c*.

With this process, the chemical structure generating device **1** according to the embodiments can efficiently generate high-quality training data appropriate as training data for machine learning and can enhance the efficiency of molecular design using a trained model produced using these training data.

The detail of this process will be described with reference to FIG. **4** to FIG. **7**.

FIG. **4** is a flowchart illustrating a process performed by the chemical structure generating device according to a first embodiment.

First, the generation function **100***a* of the processing circuit **100** starts producing a compound list (step S**100**). The generation function **100***a* of the processing circuit **100** produces a pth-generation reactant list X_p as the compound list while incrementing p, based on the zeroth-generation reactant list X_0, where p=0 in an initial state.

Subsequently, the generation function **100***a* of the processing circuit **100** produces a pth-generation product list Y_p as the product list **110***b*, based on the pth-generation reactant list X_p that is the reactant list **110***a* and the chemical reaction list **110***c* (step S**200**). That is, the generation function **100***a* of the processing circuit **100** produces the product list **110***b* including one or more compounds, based on a reactant list including one or more compounds in the reactant list **110***a* and the chemical reaction list **110***c*. The detail of the process at step S**200** will be described later with reference to FIG. **6**.

Subsequently, the control function **100***b* of the processing circuit **100** applies the pth-generation product list Y_p as a new reactant list (the (p+1)th-generation reactant list X_{p+1}) to the computer program related to the generation function **100***a* (step S**300**). That is, the control function **100***b* of the processing circuit **100** applies the product list Y_p produced at step S**200** as a new reactant list X_{p+1} to the computer program related to the generation function **100***a*.

In addition to performing such a process, the control function **100***b* of the processing circuit **100** updates the database **110** having at least one list of the reactant list **110***a* and the product list **110***b*. Here, updating the database **110** means an operation of changing elements of the database **110** by adding a new element to the database **110**, for example, an operation of inputting some elements of the product list **110***b* as new elements of a new reactant list **110***a* to the database **110**. For example, the control function **100***b* of the processing circuit **100** updates the reactant list **110***a* at step S**300**.

At least a part of the database **110** updated in this way is used, for example, as training data in machine learning. As an example, the processing circuit **100** has a not-illustrated training data producing function to produce training data in machine learning based on the updated reactant list **110***a*. As an example, the not-illustrated training data producing function of the processing circuit **100** may associate each of the elements of the updated reactant list **110***a* with the physical property value of the element to create supervised data in which a chemical structure is associated with the physical property value of the chemical structure, and the supervised data may be used as training data in machine learning. Machine learning is performed using such training data to produce a trained model, whereby the physical property value of an unknown chemical structure can be predicted and thus, for example, a chemical structure having a desired physical property value can be searched for. As another example, the not-illustrated training data producing function of the processing circuit **100** may create unsupervised data based on the updated reactant list **110***a*, and the unsupervised data may be used as training data in machine learning.

Subsequently, the control function **100***b* of the processing circuit **100** increments the value of p by 1 (step S**400**). Here, when the value of p is greater than a preset threshold p_{max} (Yes at step S**500**), the process ends. That is, when the iterative generation number p exceeds the preset threshold p_{max}, the processing circuit **100** having the control function **100***b* determines that the termination condition of the product list creating process is satisfied and terminates the process. On the other hand, if the value of p is smaller than p_{max} (No at step S**500**), the processing circuit **100** repeats the process from step S**200** to step S**400**. Specifically, the control function **100***b* of the processing circuit **100** causes the computer program related to the generation function **100***a* to produce a new product list based on the new reactant list at step S**300** and the chemical reaction list in the chemical reaction list **110***c*. The termination condition of the product list creating process is not limited to the one that the process is terminated when the iterative generation number p exceeds a certain iteration count. The processing circuit **100** may terminate the product list creating process based on another termination condition, for example, when the number of elements of the produced product list reaches a certain value.

FIG. **5** illustrates an example of compounds successively produced by such a process. FIG. **5** is a diagram illustrating a process performed by the chemical structure generating device according to the first embodiment.

A compound **10***a* and a compound **10***b* denote an example of compounds included in a zeroth-generation reactant list X_0 in an initial state, that is, p=0. With p=0, at step S**200**, the generation function **100***a* of the processing circuit **100** produces a compound **11** as a zeroth-generation product list Y_0, based on the compound **10***a* and the compound **10***b*. With p=0, at step S**300**, the control function **100***b* of the processing circuit **100** applies the zeroth-generation product list Y_0 as a first-generation reactant list X_1 to the computer program related to the generation function **100***a*. That is, the compound **11** is included in the first-generation reactant list X_1. The processing circuit **100** having the control function **100***b* stores the product list of these produced by the generation function **100***a* into the storage unit **120**. Subsequently, at step S**400**, the control function **100***b* of the processing circuit **100** increments the value of p by one to set p=1.

Subsequently, with p=1, at step S**200**, the generation function **100***a* of the processing circuit **100** produces a compound **12***a* based on the compound **10***a* and the compound **11** and a compound **12***b* based on the compound **11**, as a first-generation product list Y_1. With p=1, at step S**300**, the control function **100***b* of the processing circuit **100** applies the first-generation product list Y_1 as a second-generation reactant list X_2 to the computer program related to the generation function **100***a*. That is, the compound **12***a* and the compound **12***b* are included in the second-generation reactant list X_1. Subsequently, at step S**400**, the control function **100***b* of the processing circuit **100** increments the value of p by one to set p=2.

Subsequently, the generation function **100***a* of the processing circuit **100** produces a compound **13***a* based on the compound **10***b* and the compound **12***b* and produces a compound **14** based on the compound **10***a* and the compound **13***a*, in the same manner. By doing so, the chemical structure generating device **1** can successively produce various compounds efficiently. In FIG. **5**, the chemical reaction that produces a compound **13***b* from the compound **12***a* and the compound **11** corresponds to a chemical reaction included in the prohibitive reaction list **110***d*, and in such a case, the processing circuit **100** does not add the produced compound **13***b* to the product list.

The detail of the process at step S**200** in FIG. **4** will now be described with reference to FIG. **6**. That is, the flowchart in FIG. **6** is a flowchart illustrating the detail of the process at step S**200** in FIG. **4**.

As previously mentioned, the reactant list X_k of each generation may include the special compounds x_{null} and x_{intra} representing special operations as optional constituent elements, in addition to the compounds $x_{kl} \ldots x_{kn}$ representing reactants. These special compounds will be described later. Here, a case where the reactant list X_k of each generation does not include these special compounds will be described first.

In the following, in order to avoid complication of explanation, in FIG. **6**, a chemical reaction used for producing a product list is a chemical reaction that produces one product from two reactants.

First of all, at step S**210**, the generation function **100***a* of the processing circuit **100** starts producing the pth-generation product list Y_p in the product list **110***b*. Here, the generation function **100***a* of the processing circuit **100** starts producing the product list Y_p of each pth generation, based on the zeroth-generation reactant list X_0, the first-generation reactant list X_1, the second-generation reactant list X_2, . . . and the pth-generation reactant list X_p. Immediately after the generation function **100***a* of the processing circuit **100** starts producing the product list Y_p, the product list Y_p is an empty set. When the process is performed up to the p_0th generation, the union of the product lists $Y_0, Y_1 \ldots Y_{p0}$ produced in this way serves as data to be used as training data for machine learning. In other words, the generation function **100***a* of the processing circuit **100** produces data to be used as training data for machine learning, as the union of the produced product lists $Y_0, Y_1 \ldots Y_{p0}$.

First, at step **3220**, the generation function **100***a* of the processing circuit **100** selects one element from any one reactant list among the zeroth-generation reactant list X_0, the first-generation reactant list X_1, the second-generation reactant list X_2, . . . and the pth-generation reactant list X_p, and extracts the selected element as a first element x_m.

For example, a case of the zeroth-generation reactant list $X_0=\{x_{01}, x_{02}\}$ and the first-generation reactant list $X_1=\{x_{11}, x_{12}\}$, where x_{01}="CH_3OH", x_{02}="CH_4", x_{11}="CH_3COOH", x_{12}="C_2H_5COOH", p=1, will be discussed. In this case, for example, at step S**220**, the generation function **100***a* of the processing circuit **100** selects one element from the zeroth-generation reactant list X_0 and extracts the element x_{02}="CH_4" as a first element x_m. As another example, at step S**220**, the generation function **100***a* of the processing circuit **100** selects one element from the first-generation reactant list X_1 and extracts the element x_{12}="C_2H_5COOH" as a first element x_m.

Subsequently, at step S**230**, the generation function **100***a* of the processing circuit **100** selects one element from the pth-generation reactant list X_p and extracts the selected element as a second element x_n. For example, when p=1, at step S**230**, the generation function **100***a* of the processing circuit **100** selects one element from the first-generation reactant list X_1 and extracts the element x_{11}="CH_3COOH" as a second element x_m. In this way, at step S**220** and step S**230**, the generation function **100***a* of the processing circuit **100** extracts one or more compounds from the reactant list.

The generation function **100***a* of the processing circuit **100** repeats the process at step S**220** to step S**280** for various first elements x_m and second elements x_n.

In this iterative process, the processing circuit **100** may perform the process at step S**220** to step S**280** for all combinations of first element x_m and second element x_n or conversely may perform the process at step S**220** to step S**280** for some combinations of possible combinations of first element x_m and second element x_n.

Subsequently, at step S**240** and step S**250**, the generation function **100***a* of the processing circuit **100** selects a partial structure subjected to a chemical reaction from the selected element.

At step S**240**, the generation function **100***a* of the processing circuit **100** extracts a first partial structure y_m from the first element x_m extracted at step S**220**. For example, when the first element x_m extracted at step S**240** is "C_2H_5COOH", this structure matches a structure in the form of "R—COOH", and then at step S**240**, the generation function **100***a* of the processing circuit **100** extracts a first partial structure y_m as y_m="COOH". As another example of extraction of a partial structure, when the first element x_m extracted at step S**240** is "C_2H_5COOH", this structure matches a structure in the form of "R—H", and then at step S**240**, the generation function **100***a* of the processing circuit **100** extracts a first partial structure y_m as y_m="H".

Similarly, at step S**250**, the generation function **100***a* of the processing circuit **100** extracts a second partial structure y_n from the second element x_n extracted at step S**240**.

The generation function **100***a* of the processing circuit **100** repeats the process at step S**240** to step S**280** for various first partial structures y_m and second partial structures y_n.

Subsequently, the generation function **100***a* of the processing circuit **100** reads the chemical reaction list **110***c* from the storage unit **120** and determines whether the structure extracted at step S**240** and step S**250** is included in a chemical reaction listed in the chemical reaction list **110***c*. If the first partial structure y_m and the second partial structure y_n do not match the chemical reaction list (No at step S**260**), the process at step **S240** to step **S280** is repeated for new first partial structure y_m and second partial structure y_n. On the other hand, if the first partial structure y_m and the second partial structure y_n match the chemical reaction list (Yes at step **S260**), the process proceeds to step **S265**. For example, when the first partial structure y_m is "COOH" and the second partial structure y_n is "H", these partial structures match the chemical reaction list {"X_1—CO—X_2", {"X_1—COOH", "X_2—H"}}. In such a case, therefore, the generation function **100***a* of the processing circuit **100** determines that a new compound can be produced by the chemical reaction, and the process proceeds to step **S265**.

In this way, the generation function **100***a* of the processing circuit **100** extracts a chemical reaction in which one or more compounds extracted at steps **S220** and **S230** are reactants, from the chemical reaction list **110***c*.

Subsequently, at step **S265**, the generation function **100***a* of the processing circuit **100** reads the prohibitive reaction list **110***d* from the storage unit **120** and determines whether the first partial structure y_m extracted at step **S240** and the second partial structure y_n extracted at step **S250** match a prohibition rule. If the first partial structure y_m and the second partial structure y_m match the prohibition rule (Yes at step **S265**), the processing circuit **100** does not perform a new process for this combination of partial structures and repeats the process at step **S240** to step **S280** for new first partial structure y_m, and second partial structure y_n. For example, when the first partial structure y_m is "—COOH" and the second partial structure y_n is "—CO—CH_2—CO—", these partial structures match the prohibitive reaction list {"X_2—CO—CHX_1—CO—X_3", {"X_1—COOH", "X_2—CO—CHX_1—CO—X_3"}. In such a case, therefore, the generation function **100***a* of the processing circuit **100** determines that the new compound produced by the chemical reaction has no commercial value, and the processing circuit **100** does not perform a new process for this combination of partial structures and repeats the process at step **S240** to step **S280** for new first partial structure y_m and second partial structure y_n. On the other hand, if the first partial structure y_m and the second partial structure y_n do not match the prohibition rule (No at step **S265**), the process proceeds to step **S270**, and the resultant product is added to the product list Y_p.

In this way, the generation function **100***a* of the processing circuit **100** produces the product list Y_p further based on the prohibition rule. In the foregoing example, the prohibition rule is defined based on chemical reactions, that is, the prohibition rule is a rule that defines a chemical reaction excluded from the chemical reaction list at step **S260**. However, the embodiments are not limited thereto. That is, the prohibition rule may be defined based on products or reactants. In other words, the prohibition rule may be a rule that defines a product excluded from the product list Y_p or a reactant corresponding to the product excluded from the product list Y_p. The embodiments are not limited to the foregoing examples, and the generation function **100***a* of the processing circuit **100** may define the product list Y_p, based on a permission rule that is a rule defining a chemical reaction incorporated into the chemical reaction list, rather than the prohibition rule that is a rule defining a chemical reaction excluded from the chemical reaction list.

Subsequently, at step **S270**, the generation function **100***a* of the processing circuit **100** produces a product y_{mn} based on the chemical reaction list. For example, when the first element x_m is "C_2H_5COOH", the second element x_n is "CH_4", the first partial structure y_n is "—COOH", the second partial structure y_m is "—H", and the chemical reaction list is {"X_1—CO—X_2", {"X_1—COOH", "X_2—H"}}, the product y_{mn} produced by the generation function **100***a* of the processing circuit **100** is "C_2H_5—CO—CH_3".

Subsequently, at step **S275**, the generation function **100***a* of the processing circuit **100** determines whether the product y_{mn} produced at step **S270**, that is, the compound serving as a candidate for the product list Y_p is included in a product list previously produced, that is, a compound already produced in the previous process. If the generation function **100***a* of the processing circuit **100** determines that the product y_{mn} produced at step **S270** is included in a product list previously generated, that is, if it is determined that the product y_{mn} is a compound already produced in the previous process (Yes at step **S275**), the processing circuit **100** does not include the product y_{mn} in the pth-generation product list Y_p and repeats the process at steps **S240** to **S280** for new first partial structure y_m and second partial structure y_n.

On the other hand, if the generation function **100***a* of the processing circuit **100** determines that the product y_{mn} produced at step **S270** is not a compound already produced in the previous process (No at step **S275**), the process proceeds to step **S280**, and the processing circuit **100** adds the product y_{mn} to the pth-generation product list Y_p. In this way, the generation function **100***a* of the processing circuit **100** adds the product y_{nm} of the chemical reaction extracted at step **S260** and produced at step **S270** to the list of product Y_p made of one or more compounds.

The generation function **100***a* of the processing circuit **100** repeats the process at step **S240** to **S280** for various first partial structures y_m and second partial structures y_n while changing the first partial structure y_m and the second partial structure y_n for given first element x_m and second element x_n. The generation function **100***a* of the processing circuit **100** also repeats the process at step **S220** to step **S280** for various first elements x_m and second elements x_n.

The embodiments are not limited to the foregoing example.

In FIG. **6**, the chemical reaction used for production of a compound is a chemical reaction that produces one product from two reactants (the number of dimensions of the reaction is two). However, the embodiments are not limited thereto. As an example, the chemical reaction used for production of a compound may be a chemical reaction that produces one product from three reactants (the number of dimensions of the reaction is three), may be a chemical reaction that produces two products from one reactant, or may be a chemical reaction that produces one product from one reactant where the product is different from the reactant.

For example, a case where a chemical reaction used for production of a compound is a chemical reaction that produces one product from three reactants will be described. In such a case, instead of step **S220** to step **S230**, the generation function **100***a* of the processing circuit **100** extracts a first element x_{m1} from any one reactant list among the zeroth- to pth-generation reactant lists X_0 to X_p, extracts a second element x_{m2} from any one reactant list among the zeroth- to pth-generation reactant lists X_0 to X_p, and extracts a third element x_{m3} from the pth-generation reactant list X_p. Furthermore, instead of step **S240** to step **S250**, the generation function **100***a* of the processing circuit **100** extracts a first partial structure y_{m1} from the first element x_{m1}, extracts a second partial structure y_{m2} from the second element x_{m2}, and extracts a third partial structure y_{m3} from the third element x_{m3}. The generation function **100***a* of the processing circuit **100** performs a similar process for various first partial structures y_{m1}, second partial structures y_{m2}, and third partial structures y_{m3} for a given combination of first element x_{m1}, second element x_{m2}, and third element x_{m3}, and performs these processes for various combinations of first element x_{m1}, second element x_{m2}, and third element x_{m3}.

For example, a case where a chemical reaction used for production of a compound is a chemical reaction that produces two products from one reactant will be described. In such a case, instead of step S220 to step S230, the generation function 100a of the processing circuit 100 extracts an element x_m from the pth-generation reactant list X_p. Furthermore, instead of step S240 to step S250, the generation function 100a of the processing circuit 100 extracts a partial structure y_m from the element x_m. The generation function 100a of the processing circuit 100 acquires a chemical reaction matched with the partial structure y_m from the chemical reaction list 110c at step S260, checks a prohibition rule at step S265, and thereafter produces two products $y_{m:1}$, $y_{m;2}$ based on the chemical reaction list at step S270. At step S275, it is determined whether each of these products is a compound already produced and thereafter, if not a compound already produced, adds the product to the pth-generation product list Y_p at step S280.

These processes can be easily generalized when a chemical reaction used for production of a compound is a chemical reaction that produces r products from q reactants.

The chemical reaction used for production of a compound may be, for example, a chemical reaction that produces one product from one reactant, such as a chemical reaction that produces a product "R—Br" from a reactant "R—OH". This chemical reaction can be represented, for example, by {{"R—Br"}, {"R—OH"}}.

In the embodiment described above, the processing circuit 100 having the generation function 100a extracts a first element x_m from the zeroth-generation to pth-generation reactant lists X_0 to X_p at step S220 and extracts a second element x_n from the pth-generation reactant list X_p at step S230. However, the embodiments are not limited thereto. For example, the processing circuit 100 having the generation function 100a may extract a second element x_n from the zeroth-generation to pth-generation reactant lists X_0 to X_p also at step S230.

A case where the reactant list X_p includes a special compound x_{null} as an element in FIG. 6 will now be described. The special compound x_{null} is a special element indicating an operation of not selecting a compound, and selecting the special compound x_{null} is treated as no compound being selected.

For example, a case where the reactant list X_p is a list of n+1 elements including n elements $x_1, x_2 \ldots x_n$ that are normal compounds and the special compound x_{null}, and three elements are extracted from this list where overlapping is allowed will be discussed. When three compounds other than the special compound x_{null}, such as x_k, x_l, x_m, are extracted, the generation function 100a of the processing circuit 100 extracts a three-dimensional reaction with three reactants x_k, x_l, x_m. A case where x_{null} is extracted once and two compounds other than x_{null}, such as x_k, x_l, are extracted is treated as one compound not being selected, and the generation function 100a of the processing circuit 100 extracts a two-dimensional reaction with two reactants x_k, x_l. A case where x_{null} is extracted twice and one compound other than x_{null}, such as x_k, is extracted is treated as two compounds not being selected, and the generation function 100a of the processing circuit 100 extracts a one-dimensional reaction with one reactant x_k. In this way, by making a list of n+1 elements including n compounds $x_1, x_2 \ldots x_n$ and the special compound x_{null} as the reactant list X_p and extracting d elements from this list, where overlapping is allowed, the generation function 100a of the processing circuit 100 can exhaustively count up chemical reactions in which the number of reactants is 1 to d, that is, the dimensions of reaction are equal to or smaller than d. That is, the reactant list X_p includes an operation of not selecting a compound as the element x_{null}, and the generation function 100a of the processing circuit 100 extracts the element x_{null} from the reactor list X_p instead of extracting a compound, whereby compounds obtained by a chemical reaction involving a smaller number of reactants than when the element x_{null} is not extracted can be added to the product list Y_p.

The reactant list X_p may include an operation of converting a structure of a certain compound into a compound different from the original compound, as an element. Here, an example of the "operation of converting a structure of a certain compound into a compound different from the original compound" is an operation such as intramolecular reaction, isomerization, and substitution of a functional group.

Such an operation will be described with reference to FIG. 7, taking an intramolecular reaction as an example. FIG. 7 is a diagram illustrating a process performed by the chemical structure generating device according to the first embodiment.

First, a normal chemical reaction that is not an intramolecular reaction will be described. A case where the compound x_0 included in the reactant list X_p is a compound 15 illustrated in FIG. 7 will be discussed. Here, in the case of a normal chemical reaction that is not an intramolecular reaction, as illustrated in the middle section of FIG. 7, two molecules of the compound 15 come together to produce a compound 16. In such a case, the reactants of the chemical reaction are represented as {"compound 15", "compound 15"}, the product of the chemical reaction is represented as "compound 16", and the chemical reaction is represented as {{"compound 16"}, {{"compound 15", "compound 15"}.

Next, a case where an intramolecular reaction is included will be discussed. When an intramolecular reaction exists, as illustrated in the lower section of FIG. 7, an intramolecular reaction occurs in one molecule of the compound 15 to produce a compound 17. In such a case, although the reactant of the chemical reaction can be represented as "compound 15", the product of the chemical reaction can be represented as "compound 17", and the chemical reaction can be represented as {"compound 17", "compound 15"}, the processing circuit 100 having the generation function 100a may treat the operation x_{intra} of performing an intramolecular reaction as an element included in the reactant list X_p. Specifically, the processing circuit 100 having the generation function 100a represents the reactant of the reaction as {"compound 15", x_{intra}} using the special element corresponding to the operation x_{intra} of performing an intramolecular reaction, the product of the chemical reaction as "compound 17", and the chemical reaction as {{"compound 17"}, {"compound 15", x_{intra}}} and then performs the process illustrated in FIG. 6. The processing circuit 100 having the generation function 100a treats the operation x_{intra} of performing an intramolecular reaction as an element included in the reactant list X_p to add a compound obtained by performing an intramolecular reaction on any compound in the reactant list to the product list Y_p. As indicated by this example, by treating the operation on a molecule as an element included in a set of molecule candidates, the processing circuit 100 having the generation function 100a can systematically generate molecule candidates obtained by an operation such as an intramolecular reaction.

The product y_{nm} produced by the generation function **100***a* of the processing circuit **100** at step S**270** is typically some kind of chemical structure (compound). However, the embodiments are not limited thereto. For example, at step S**270**, the generation function **100***a* of the processing circuit **100** may generate the special compound x_{null} or x_{intra} as a product y_{nm}. For example, when all the elements selected as reactants are x_{null}, the generation function **100***a* of the processing circuit **100** generates x_{null} as a product at step S**270**. When all the elements selected as reactants are x_{intra}, the generation function **100***a* of the processing circuit **100** generates x_{intra} as a product at step S270. When only a special compound is selected as a reactant, the process of step S**240**, step S**250**, step S**260**, step S**265**, step S**275**, or the like in FIG. **6** is not necessarily performed.

As described above, in the chemical structure generating device **1** according to the first embodiment, the generation function **100***a* of the processing circuit **100** produces a product list including one or more compounds based on a reactant list including one or more compounds and a chemical reaction list, the control function **100***b* applies the product list as a new reactant list and allows the generator to produce a new product list based on the new reactant list and the chemical reaction list. This process enables efficient generation of molecular structures appropriate for use as training data for machine learning, molecular structures having commercial value, and molecular structures that can be actually manufactured.

Second Embodiment

In a second embodiment, a chemical structure generating device will be described that generates molecular structures having commercial value even more efficiently by assigning numerical information to each of the compounds serving as candidates and performing control, as a method of efficiently generating molecular structures having commercial value.

In the second embodiment, the chemical structure generating device **1** assigns numerical information to each of the compounds serving as candidates based on the numerical information list **110***e* and performs control based on the numerical information. As used herein the numerical information assigned to each of the compounds is, for example, information indicating a cost necessary for producing the compound.

In a typical case, the numerical information is assigned to each compound.

For example, a case where the pth-generation reactant list X_p includes compounds $x_1, x_2, \ldots x_n$, and the costs for producing the compounds $x_1, x_2, \ldots, x_n$ are $w_1, w_2, \ldots, w_n$, respectively, will be discussed. In this case, the elements in the numerical information list W_p corresponding to the pth-generation reactant list are $w_1, w_2, \ldots w_n$.

The processing circuit **100** having the generation function **100***a* may integrate the reactant list **110***a* and the numerical information list **110***e* to produce one compound list $X'_p=\{x_1(w_1), x_2(w_2), \ldots x_n(w_n)\}$ in which numerical information is set as an attribute for each compound.

The generation function **100***a* of the processing circuit **100** calculates the numerical information for each compound. Specifically, the processing circuit **100** having the generation function **100***a* provides known numerical information as numerical information for a compound whose numerical information is known and, for numerical information of any other compound, recursively defines numerical information based on numerical information of reactants of the compound, based on a synthesis pathway of the compound, until reaching a compound with known numerical information.

Such a process will be specifically described with reference to FIG. **8** and FIG. **9**. FIG. **8** and FIG. **9** are flowcharts illustrating a process performed by the chemical structure generating device according to the second embodiment. FIG. **8** is a flowchart illustrating the entire process, and FIG. **9** is a flowchart illustrating the process at step S**200**A in FIG. **8**. A description of the process in FIG. **8** and FIG. **9** already explained in the first embodiment will not be repeated.

First of all, the generation function **100***a* of the processing circuit **100** starts producing a compound list (step S**100**A). The generation function **100***a* of the processing circuit **100** produces a pth-generation reactant list X_p and a numerical information list W_p corresponding to the reactants of the pth-generation reactant list X_p, as a compound list and a numerical information list (a list of cost information), respectively, while incrementing p, based on the zeroth-generation reactant list X_0 and the numerical information list W_0 corresponding to the reactants, where p=0 in an initial state. As an example, the processing circuit **100** having the generation function **100***a* initially produces a reactant list $X_0=\{x_1, x_2, \ldots x_n\}$ and a list of information such as numerical information (cost information) $W_0=\{w_1, w_2, \ldots, w_n\}$ and initially produces a reactant list $X_0'=\{x_1(w_1), x_2(w_2), \ldots, x_n(w_n)\}$ in which numerical information is associated with a reactant, based on these lists, to start producing a compound list.

Subsequently, the generation function **100***a* of the processing circuit **100** produces a pth-generation product list Y_p as a product list **110***b*, based on the pth-generation reactant list X_p that is the reactant list **110***a* and the chemical reaction list **110***c* that is a list of chemical reactions. In addition to this process, the generation function **100***a* of the processing circuit **100** also calculates a numerical information list W_p corresponding to the products of the pth-generation product list Y_p. That is, the processing circuit **100** having the generation function **100***a* produces the pth-generation product list Y_p in such a manner that numerical information is associated with each product (step S**200**A). The detail of the process at step S**200**A will be described later with reference to FIG. **9**.

Subsequently, the control function **100***b* of the processing circuit **100** applies the pth-generation product list Y_p and the numerical information list W_p corresponding to the products, as a new reactant list (the (p+1)th-generation reactant list X_{p+1}) and a new numerical information list W_{p+1}, respectively, to the computer program related to the generation function **100***a* (step S**300**A). In this way, the processing circuit **100** having the control function **100***b* applies the pth-generation product list Y_p in which numerical information is associated with each product, as a new reactant list (the (p+1)th-generation reactant list X_{p+1}), to the computer program related to the generation function **100***a*.

Subsequently, the control function **100***b* of the processing circuit **100** increments the value of p by 1 (step S**400**). Subsequently, the control function **100***b* of the processing circuit **100** determines whether the new reactant list X_p is an empty set. In the second embodiment, the processing circuit **100** having the generation function **100***a* performs control using the numerical information list W_p and restricts compounds to be produced. Therefore, as p increases, the number of elements of a new reactant list X_p naturally decreases. The processing circuit **100** having the control function **100***b* therefore can set an end point of the process involving p, at p in which a new reactant list X_p is an empty set.

That is, when a new reactant list X_p is an empty set (Yes at step **S500**A), the process ends, and if a new reactant list X_p is not an empty set (No at step **S500**A), the process returns to step **S200**A. Such a stop condition of the process enables efficient generation of a commercially valuable compound, for example, whose cost is smaller than a given threshold.

The detail of the process at step **S200**A will now be described with reference to FIG. **9**. In FIG. **9**, the numerical information list W_p corresponding to the reactants is information representing the costs as a product of the reactants.

In FIG. **9**, a chemical reaction used in production of a product list is a chemical reaction that produces one product from two reactants, in the same manner as in FIG. **6**. However, as already mentioned, the embodiment can be applied similarly even when the number of reactants or products is different from the above.

First of all, at step **S210**A, the generation function **100***a* of the processing circuit **100** starts producing a pth-generation product list Y_p in the product list **110***b*. Immediately after the generation function **100***a* of the processing circuit **100** starts producing the product list Y_p, the product list Y_p is an empty set.

At step **S220**A, the generation function **100***a* of the processing circuit **100** selects one element from any one reactant list among the zeroth-generation reactant list X_0, the first-generation reactant list X_1, the second-generation reactant list X_2, . . . and the pth generation-reactant list X_p and extracts the selected element as a first element x_m, in the same manner as in the first embodiment.

Subsequently, at step **S230**A, the generation function **100***a* of the processing circuit **100** selects one element from the pth-generation reactant list X_p and extracts the selected element as a second element x_n. In this way, at step **S220** and step **S230**, the generation function **100***a* of the processing circuit **100** extracts one or more compounds from the reactant list.

The generation function **100***a* of the processing circuit **100** repeats the process at step **S220**A to step **S280**A for various first elements x_m and second elements x_n.

Subsequently, at step **S231**, the generation function **100***a* of the processing circuit **100** calculates the reactant cost (the sum of raw material costs) for a reaction including the first element x_m selected at step **S220**A and the second element x_n selected at step **S230**A. Here, if the product cost, which is the sum of the reactant cost (the sum of raw material costs) and the cost for performing a chemical reaction, exceeds a permissible cost at a point of time of the sum of raw material costs, the cost for a chemical reaction need not be considered and the product cost exceeds the maximum permissible cost. The processing circuit **100** having the generation function **100***a* therefore performs a pruning process at step **S232** before entering a loop for partial structures.

Specifically, if the reactant cost, for example, the sum w_m+w_n of the cost w_m for the first element x_m selected at step **S220**A and the cost w_n for the second element x_n selected at step **S230**A exceeds a permissible cost (No at step **S232**), the processing circuit **100** having the generation function **100***a* determines that the first element x_m and the second element x_n selected at step **S220**A and step **S230** are not appropriate elements and repeats the process at step **S220**A to step **S280**A for new first element x_m and second element x_n. On the other hand, if the reactant cost, for example, the sum w_m+w_n of the cost w_m for the first element x_n selected at step **S220**A and the cost w_n for the second element x_n selected at step **S230**A does not exceed a permissible cost (Yes at step **S232**), the process proceeds to step **S240**.

Subsequently, at step **S240** and step **3250**, the generation function **100***a* of the processing circuit **100** selects a partial structure subjected to a chemical reaction from the selected element. For example, at step **S240**, the generation function **100***a* of the processing circuit **100** extracts a first partial structure y_m from the first element x_m extracted at step **S220**. Similarly, at step **S240**, the generation function **100***a* of the processing circuit **100** extracts a second partial structure y_n from the second element x_n extracted at step **S230**.

The generation function **100***a* of the processing circuit **100** repeats the process at step **S240** to step **S280**A for various first partial structures y_m and second partial structures y_n.

Subsequently, the generation function **100***a* of the processing circuit **100** reads the chemical reaction list **110***c* from the storage unit **120** and determines whether the structures extracted at step **S240** and step **S250** are included in a chemical reaction listed in the chemical reaction list **110***c*. If the first partial structure y_m and the second partial structure y_m do not match the chemical reaction list (No at step **S260**), the process at step **S240** to step **S280**A is repeated for new first partial structure y_m and second partial structure y_n. On the other hand, if the first partial structure y_m and the second partial structure y_n match the chemical reaction list (Yes at step **S260**), the process proceeds to step **S270**.

In this way, the generation function **100***a* of the processing circuit **100** extracts a chemical reaction in which one or more compounds extracted at steps **S220**A and **S230**A are reactants, from the chemical reaction list **110***c*.

Subsequently, at step **S270**, the generation function **100***a* of the processing circuit **100** produces a product y_{mn} based on the chemical reaction list.

Subsequently, at step **S275**, the generation function **100***a* of the processing circuit **100** determines whether the product y_{mn} produced at step **S270**, that is, the compound serving as a candidate for the product list Y_p is included in a product list previously produced, that is, whether it is a compound already produced in the previous process. If the generation function **100***a* of the processing circuit **100** determines that the product y_{mn} produced at step **S270** is included in a product list previously produced, that is, if it is determined that the product y_{mn} is a compound already produced in the previous process (Yes at step **S275**), the processing circuit **100** does not include the product y_{mn} in the pth-generation product list Y_p and repeats the process at steps **S240** to **S280** for new first partial structure y_m and second partial structure y_n.

On the other hand, if the generation function **100***a* of the processing circuit **100** determines that the product y_{mn} produced at step **S270** is not a compound already produced in the previous process (No at step **S275**), the process proceeds to step **S271**A.

At step **S271**A, the processing circuit **100** having the generation function **100***a* calculates the production cost w_{mn} for the product y_{mn} produced at step **S270**. Here, the production cost w_{mn} is the sum of the sum w_m+w_n of reactant costs and the cost z_{mn} for a chemical reaction. Specifically, the processing circuit **100** having the generation function **100***a* produces numerical information corresponding to each of compounds y_{mn} serving as candidates for the product list Y_p, using the sum w_m+w_n of the costs for reactants that are raw materials of a compound serving as a candidate and the cost z_{mn} for a chemical reaction $\{y_{mn}, \{x_m, x_n\}\}$ producing the compound y_{mn} serving as a candidate from the reactants. Here, the processing circuit **100** having the generation function **100**a acquires the cost z_{mn} for the chemical reaction from the chemical reaction list **110**c. The chemical structure generating device **1** stores the chemical reaction and the cost for the chemical reaction into the storage unit **120** in association with the chemical reaction list **110**c.

Subsequently, at step S**272**, the generation function **100**a of the processing circuit **100** determines whether the production cost w_{mn} for the product y_{mm} calculated at step S**271**A is equal to or lower than the maximum permissible cost c_{max}. If the production cost w_{mn} is not equal to or lower than the maximum permissible cost c_{max} (No at step S272), the product y_{mn} is not added to the product list Y_p, and the process at step S**240** to step S**280**A is repeated for new first partial structure y_m and second partial structure y_n. On the other hand, if the production cost w_{mn} is equal to or lower than the maximum permissible cost c_{max} (Yes at step S272), the process proceeds to step S**280**A.

In this way, when producing a compound y_{mn} serving as a candidate for the product list Y_p, the processing circuit **100** having the generation function **100**a calculates the production cost w_{mn} that is numerical information corresponding to each compound y_{mn} serving as a candidate at step S**271**A and determines whether to include the compound y_{mn} serving as a candidate in the product list Y_p, based on the calculated production cost w_{mn} that is numerical information at step S**272**.

That is, each of the reactants included in the reactant list X_p is associated with information representing the cost of the reactant. The processing circuit **100** having the generation function **100**a calculates the cost w_{mn} of the compound y_{mn} serving as a candidate for the product list Y_p, based on the information representing the cost of the reactant at step S**271**A, compares the calculated cost w_{mn} of the compound y_{mn} serving as a candidate for the product list Y_p with the maximum permissible cost c_{max} that is a preset threshold at step S**272**. If the cost w_{mn} exceeds the maximum permissible cost c_{max}, which is a threshold (No at step S**272**), the processing circuit **100** does not include the compound y_{mn} serving as a candidate in the product list Y_p.

At step S**280**A, the processing circuit **100** adds the product y_{mn} to the pth-generation product list Y_p and adds the production cost w_{mn} of the product y_{mn} to the pth-generation numerical information list W_p. In this way, the generation function **100**a of the processing circuit **100** adds a product to the product list Y_p including one or more compounds.

The generation function **100**a of the processing circuit **100** repeats the process at step S**240** to S**280**A for various first partial structures y_m and second partial structures y_n while changing the first partial structure y_m and the second partial structure y_n for given first element x_m and second element x_n. The generation function **100**a of the processing circuit **100** also repeats the process at step S**220**A to step S**280**A for various first elements x_m and second elements x_n.

The embodiments are not limited to the foregoing examples.

In the embodiment described above, the numerical information list W_p corresponding to each compound is information representing the cost of the compound. However, the embodiments are not limited thereto. For example, the numerical information list W_p corresponding to each compound may be a score defined for each compound. As an example, the processing circuit **100** having the generation function **100**a may determine comprehensive commercial usability for each compound, based on not only mere prices of raw materials but also various factors such as technical level of difficulty in handling, availability of raw materials, and market demands to set a score for each compound, and may preferentially add the one having a high score or exceeding a certain threshold to the next-generation reactant list. Here, the score set for each compound may be such a score that a higher score indicates a more desirable compound, or conversely may be such a score that a lower score indicates a more desirable compound. As an example of the numerical information list W_p corresponding to each compound, information on physical properties of the compound, such as molecular weight, vapor pressure, boiling point, melting point, dipole moment, and oil-water partition coefficient (for example, log P), may be used.

In the embodiment described above, when producing a compound serving as a candidate for the product list Y_p, the processing circuit **100** having the generation function **100**a further calculates numerical information corresponding to each compound serving as a candidate and determines whether to include the compound serving as a candidate in the product list Y_p, based on the calculated numerical information. However, the embodiments are not limited thereto. For example, when producing a compound serving as a candidate for the product list Y_p, the processing circuit **100** having the generation function **100**a may further calculate information that is not numerical information corresponding to each compound serving as a candidate and may determine whether to include the compound serving as a candidate in the product list Y_p, based on the calculated information. Examples of the information that is not numerical information include information as to whether a particular substituent is included and information such as hazardous materials classification by a public period. In such a case, for example, the processing circuit **100** having the generation function **100**a excludes, from the product list Y_p, a compound serving as a candidate for the product list that includes a particular substituent or is classified in a particular hazardous material class.

In the iterative process for the first element x_m and the second element x_n in the flowchart in FIG. **9** described above, selection of the first element x_m and the second element x_n is performed successively, irrespective of the content of the numerical information list W_p of the compounds. However, the embodiments are not limited thereto. The processing circuit **100** having the generation function **100**a may search for a first element x_m and a second element x_n in accordance with a priority order based on the numerical information of each compound, for example, in order of increasing cost or in order of decreasing score and may generate a candidate y_{mn} for the product list Y_p.

In the embodiments, the cost of a compound is not limited to information merely indicating the price of the compound and may include a variety of costs other than price, such as the difficulty level of safety control and the availability or stability of raw materials.

The numerical information in the embodiments is not limited to numerical information in the form of scalar quantity but may be numerical information in the form of vector quantity or tensor quantity, if necessary.

In the embodiment described above, the processing circuit **100** successively produces a pth-generation product list Y_p based on the zeroth-generation reactant list X_0. That is, in the embodiment described above, one reactant list Y_p is produced based on one initial list named the zeroth-generation reactant list X_0. However, the embodiments are not limited thereto. A plurality of reactant lists respectively corresponding to a plurality of initial lists may be produced based on the initial lists.

For example, a case where when a certain compound (this compound itself is not necessarily a target to be computed by the present compound generating device) requires two or more raw material groups (for example, group A, group B, group C . . .), respective compound lists of the raw material groups (a compound list of group A, a compound list of group B, a compound list of group C . . .) are produced within a permissible cost will be discussed.

In this case, for example, the generation function **100***a* of the processing circuit **100** produces a pth-generation compound list $Y_{p,A}$ for compounds in group A based on the zeroth-generation reactant list $X_{0,A}$ for compounds in group A, produces a pth-generation compound list $Y_{p,B}$ for compounds in group B based on the zeroth-generation reactant list $X_{0,B}$ for compounds in group B, and produces a pth-generation compound list Y_p, c for compounds in group C based on the zeroth-generation reactant list $X_{0,c}$ for compounds in group C.

First, a first case where a permissible cost is set for each of the raw material groups will be discussed. In such a case, the generation function **100***a* of the processing circuit **100** may perform the process already explained for each of the raw material groups by setting a permissible cost set for each raw material group as the maximum permissible cost c_{max} at step S**272** in FIG. **9**. For example, a case where compound lists of two kinds of raw material groups, namely, group A and group B are produced will be discussed, where the entire permissible cost c is given and the ratio between the permissible cost for group A and the permissible cost for group B is set to 8:2. In this case, the processing circuit **100** performs the process for each group by setting a permissible cost c_{max}=0.8 c for group A and a permissible cost c_{max}=0.2 c for group B.

Next, a second case where the permissible cost is set for the whole of the raw material groups, rather than each of the raw material groups, will be discussed. In such a case, the generation function **100***a* of the processing circuit **100** performs the respective processes concurrently or alternately for the raw material groups, and then determines whether the production cost of a product is equal to or lower than the maximum permissible cost by referring to information on the permissible cost of the compound list produced in the other raw material group at the process at step S**272** in FIG. **9**. As an example, in the process at step S**272** in the process of producing a compound list of group A, if the sum of the production cost of a product produced at present and the minimum value of the production cost in the compound list of group B exceeds the permissible cost of the whole, the generation function **100***a* of the processing circuit **100** does not add the product to the product list.

By performing the process described above, the processing circuit **100** can produce compound lists for two or more raw material groups simultaneously and concurrently.

In this way, in the second embodiment, the processing circuit **100** assigns numerical information to each of the compounds serving as candidates and performs control of chemical structure generation. This configuration enables even more efficient generation of molecular structures having commercial value and suitable for use as training data for machine learning.

In this way, the embodiments of the present invention can efficiently generate molecular structures suitable for use as training data for machine learning.

The embodiments of the present invention are provided only by way of example and susceptible to various modifications and changes.

For the foregoing embodiments, the following note is disclosed as selective features of the present invention.

Note

A data processing system comprising:

a database having at least one list of a reactant list including one or more compounds and a product list including one or more compounds; and

a chemical structure generating device,

the chemical structure generating device including

a generator configured to produce the product list based on the reactant list and a chemical reaction list, and

a controller configured to apply the product list as a new reactant list to the generator, update the database, and allow the generator to produce a new product list based on the new reactant list and the chemical reaction list.

According to the present invention, molecular structures suitable for use as training data for machine learning can be efficiently generated.

The methods, processes, and/or operations described herein may be performed by code or instructions to be executed by a computer, processor, controller, or other signal processing device. The computer, processor, controller, or other signal processing device may be those described herein or one in addition to the elements described herein. Because the algorithms that form the basis of the methods (or operations of the computer, processor, controller, or other signal processing device) are described in detail, the code or instructions for implementing the operations of the method embodiments may transform the computer, processor, controller, or other signal processing device into a special-purpose processor for performing the methods herein.

Also, another embodiment may include a computer-readable medium, e.g., a non-transitory computer-readable medium, for storing the code or instructions described above. The computer-readable medium may be a volatile or non-volatile memory or other storage device, which may be removably or fixedly coupled to the computer, processor, controller, or other signal processing device which is to execute the code or instructions for performing the method embodiments or operations of the apparatus embodiments herein.

Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

What is claimed is:

1. A chemical structure generating device comprising:

a processing circuit, including:

a generator configured to produce a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list; and

a controller configured to apply the product list as a new reactant list to the generator, update a database having at least one list of the reactant list and the product list, and allow the generator to produce a new product list based on the new reactant list and the chemical reaction list,

wherein numerical information is associated with each of the reactants, the numerical information associated with each of the reactants is information representing a cost of each of the reactants as a product,

the generator produces numerical information corresponding to each of the compounds serving as the candidates, using a sum of a cost of the reactant that is a raw material of the compound serving as the candidate and a cost for a chemical reaction that produces the compound serving as the candidate from the reactant,

information representing a cost of each of the reactants is associated with the reactant,

the generator calculates a cost of a compound serving as a candidate for the product list, based on the information representing the cost of the reactant, and compares the calculated cost of the compound serving as the candidate for the product list with a preset threshold, wherein when the cost exceeds the threshold, the compound serving as the candidate is not included in the product list, and

the processing circuit is configured to produce compound lists for two or more raw material groups simultaneously and concurrently.

2. The chemical structure generating device according to claim **1**, wherein the generator extracts one or more compounds from the reactant list, extracts a chemical reaction in which the one or more compounds are reactants, from the chemical reaction list, and adds a product of the chemical reaction to the product list including the one or more compounds.

3. The chemical structure generating device according to claim **1**, wherein the generator produces the product list, further based on a prohibition rule that is a rule defining a product excluded from the product list to be produced or a chemical reaction excluded from the chemical reaction list.

4. The chemical structure generating device according to claim **1**, wherein the generator determines whether a compound serving as a candidate for the product list is included in a product list previously produced, and when determining that the compound serving as the candidate is included in the product list previously produced, the generator does not include the compound serving as the candidate in the product list.

5. The chemical structure generating device according to claim **1**, wherein when producing compounds serving as candidates for the product list, the generator further calculates information corresponding to each of the compounds serving as the candidates and determines whether to include the compounds serving as the candidates in the product list, based on the calculated information.

6. The chemical structure generating device according to claim **1**, wherein numerical information is associated with each of the reactants, and

the generator generates a candidate for the product list in accordance with a priority order based on the numerical information.

7. The chemical structure generating device according to claim **1**, wherein the reactant list includes an operation of not selecting a compound, as an element, and

the generator extracts the element from the reactant list instead of extracting a compound and adds a compound obtained by a chemical reaction involving a smaller number of reactants than when the element is not extracted, to the product list.

8. The chemical structure generating device according to claim **1**, wherein the reactant list includes an operation of converting a structure of a certain compound into a compound different from the certain compound, as an element.

9. The chemical structure generating device according to claim **1**, wherein the generator treats an operation of performing an intramolecular reaction as an element included in the reactant list and adds a compound obtained by performing an intramolecular reaction on any compound in the reactant list, to the product list.

10. The chemical structure generating device according to claim **1**, wherein the controller stores the product list produced by the generator into a storage unit as the database.

11. The chemical structure generating device according to claim **1**, wherein at least a part of the database updated is used as training data in machine learning.

12. A non-transitory computer-readable medium storing a chemical structure generating program for causing a computer to perform:

a process using a processing circuit having a generator for producing a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list;

a process using the processing circuit having a controller for applying the product list as a new reactant list to the generator;

a process using the controller for updating a database having at least one list of the reactant list and the product list;

a process using the controller to allow producing a new product list by the generator based on the new reactant list and the chemical reaction list;

a process of associating numerical information with each of the reactants, wherein the numerical information associated with each of the reactants is information representing a cost of each of the reactants as a product;

a process using the generator for producing numerical information corresponding to each of the compounds serving as the candidates, using a sum of a cost of the reactant that is a raw material of the compound serving as the candidate and a cost for a chemical reaction that produces the compound serving as the candidate from the reactant, wherein information representing a cost of each of the reactants is associated with the reactant;

a process using the generator for calculating a cost of a compound serving as a candidate for the product list, based on the information representing the cost of the reactant, and comparing the calculated cost of the compound serving as the candidate for the product list with a preset threshold, wherein when the cost exceeds the threshold, the compound serving as the candidate is not included in the product list; and

a process using the processing circuit for producing compound lists for two or more raw material groups simultaneously and concurrently.

13. A chemical structure generating method, the chemical structure generating method comprising:

producing using a generator device of a processing circuit a product list including one or more compounds, based on a reactant list including one or more compounds and a chemical reaction list;

applying using a controller device of a processing circuit the product list as a new reactant list to the generator device;

updating using the controller device a database having at least one list of the reactant list and the product list;

using the controller to allow producing a new product list by the generator device based on the new reactant list and the chemical reaction list;

associating numerical information with each of the reactants, wherein the numerical information associated with each of the reactants is information representing a cost of each of the reactants as a product;

producing using the generator device numerical information corresponding to each of the compounds serving as the candidates, using a sum of a cost of the reactant that is a raw material of the compound serving as the candidate and a cost for a chemical reaction that produces the compound serving as the candidate from the reactant, and wherein information representing a cost of each of the reactants is associated with the reactant;

calculating using the generator a cost of a compound serving as a candidate for the product list, based on the information representing the cost of the reactant, and comparing the calculated cost of the compound serving as the candidate for the product list with a preset threshold, wherein when the cost exceeds the threshold, the compound serving as the candidate is not included in the product list; and

producing with the processing circuit compound lists for two or more raw material groups simultaneously and concurrently.

14. The non-transitory computer-readable medium of claim **12**, wherein the non-transitory computer-readable medium is a random access memory (RAM), a read-only memory (ROM), a flash memory, or a hard disk.

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